

Managing toxicity in an esophageal cancer patient under first-line treatment with chemotherapy + immunotherapy

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61 year old male.

Previous medical history: hypertension.

Former smoker.

Consults at the Digestive Clinic after dysphagia for solid foods.



Gastroscopy: Tumoral infiltration of the mucosa in the distal esophagus.

Pathology report: Well differentiated **adenocarcinoma**, MMRp, **CPS = 18**, HER2-, claudin18.2-.

CT scan: primary tumor in the distal esophagus without cardinal involvement. Paraesophagic right distal adenopathy and 2 gastrophpeatic adenopathies. Two more **adenopathies** are observed **in the medial retroperitoneum**.

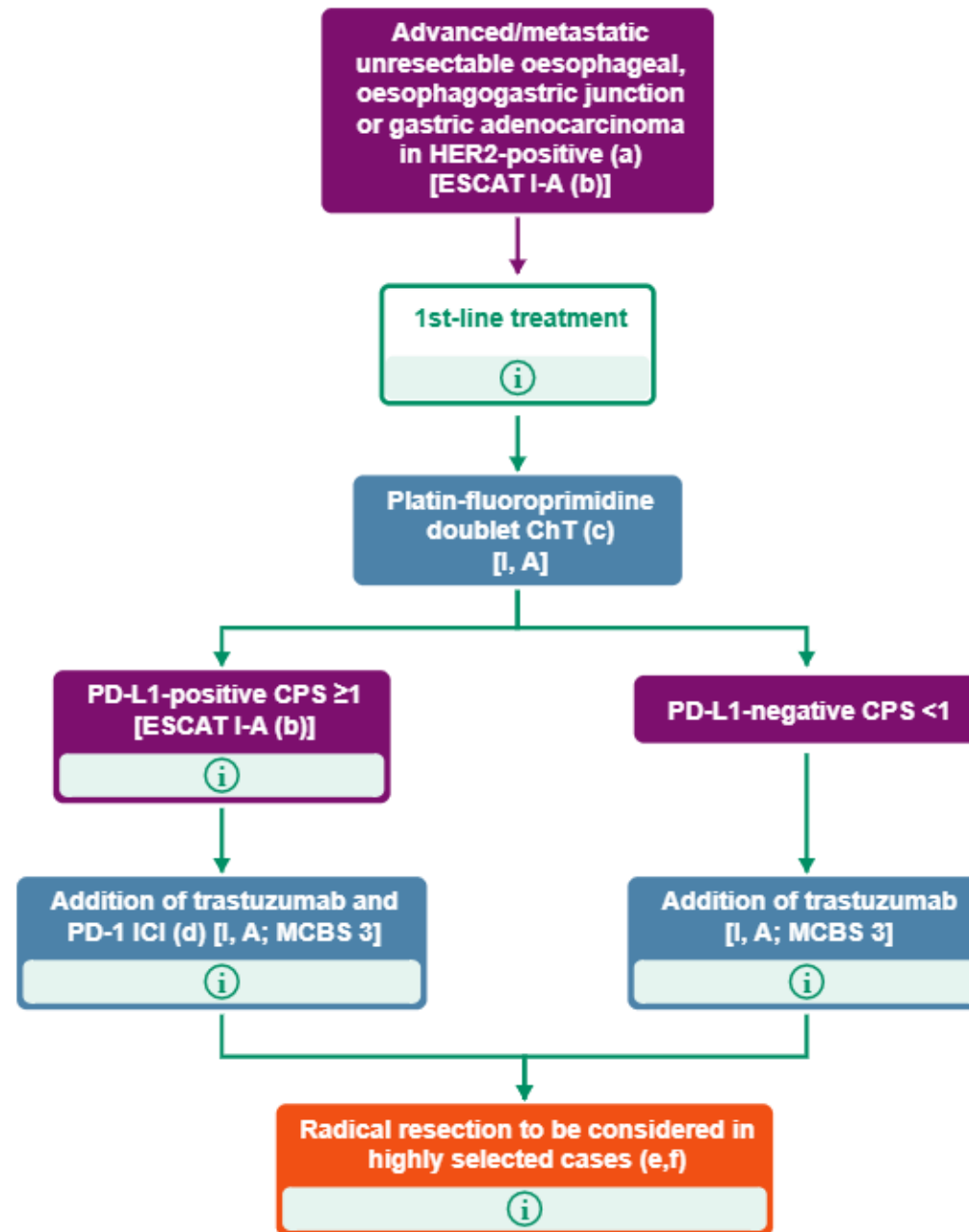
PET-CT scan: distal esophagus neoplasia. Locoregional adenopathies. **Retroperitoneal** and paraaortic **adenopathies**.



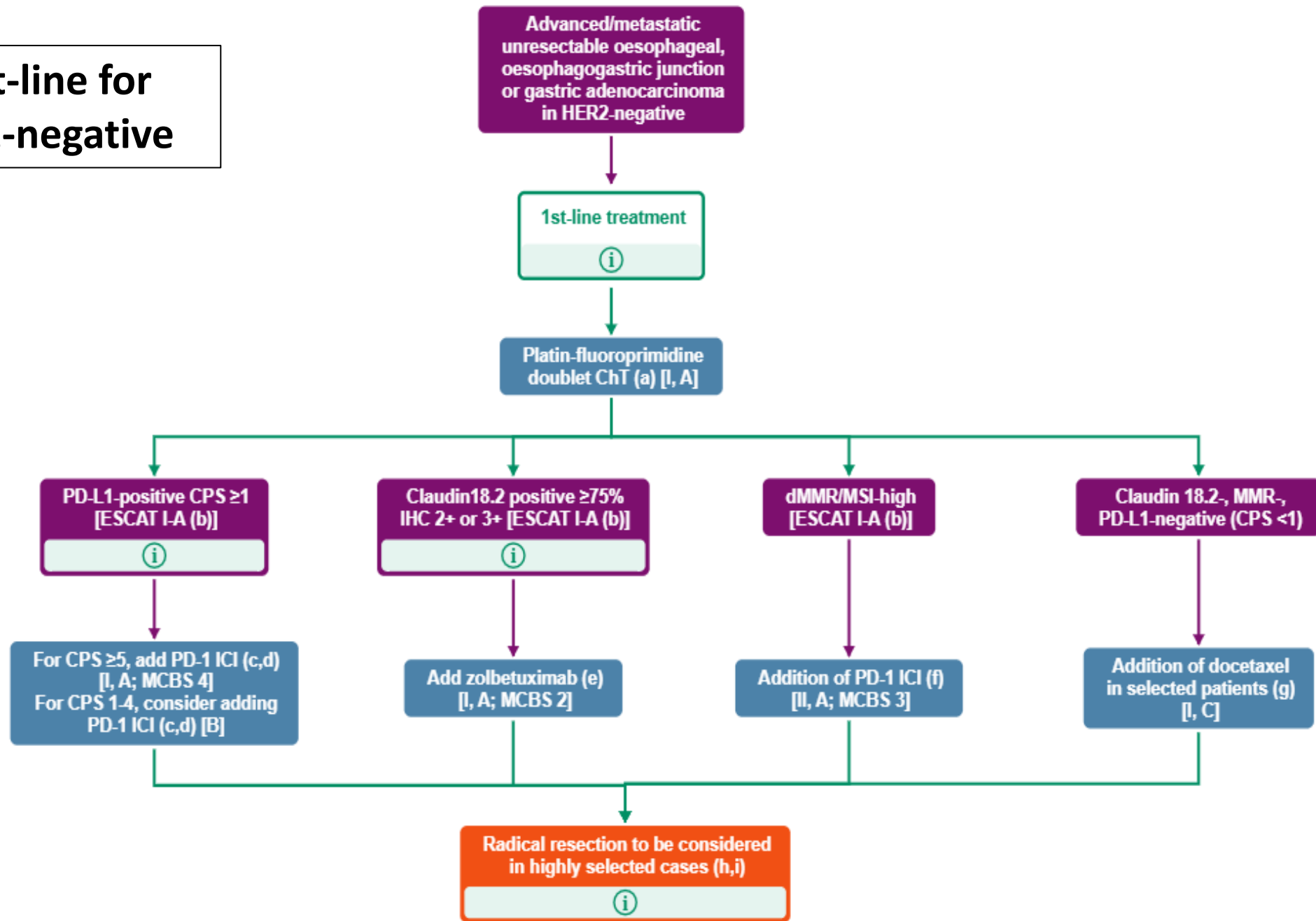
61 year old previously healthy male diagnosed with **MMRp, CPS 18, HER2-negative, claudin18.2-negative stage IV** (non-locoregional lymph nodes) **adenocarcinoma** of the **distal esophagus**

1st line
systemic therapy

First-line for HER2-positive



First-line for HER2-negative



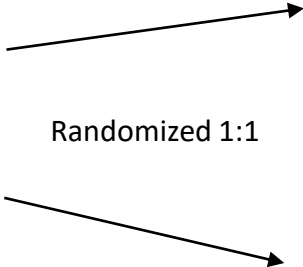
CheckMate 649

Multicentre, randomised, open-label, phase 3 trial.

Stratified by tumour cell PD-L1 status ($\geq 1\%$ vs $< 1\%$ or indeterminate), region (Asia vs USA and Canada vs rest of world), ECOG performance status (0 vs 1), and type of chemotherapy (XELOX vs FOLFOX).

Outcomes: Dual primary outcomes (**OS and PFS**) in **the PDL1 CPS ≥ 5** . Hierarchically tested secondary endpoints (OS in CPS ≥ 1 and all patients)

Adult patients with with previously untreated, unresectable advanced or metastatic gastric, GEJ, or oesophageal adenocarcinoma, regardless of PD-L1 expression
(n = 1581)



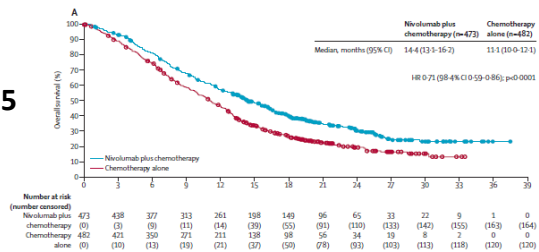
Nivolumab (360 mg Q3W or 240 mg Q2W) + Investigator’s choice of chemotherapy (CAPOX or FOLFOX)
(n=789)

Investigator’s choice of chemotherapy (CAPOX or FOLFOX)
(n=792)

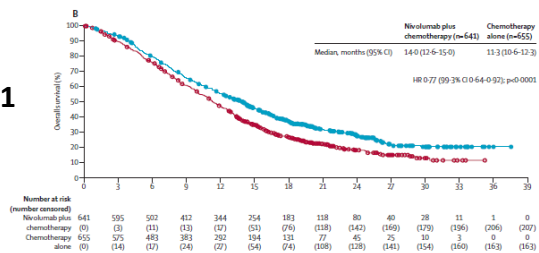
	Patients with a PD-L1 CPS of five or more		All randomly assigned patients	
	Nivolumab plus chemotherapy (n=473)	Chemotherapy alone (n=482)	Nivolumab plus chemotherapy (n=789)	Chemotherapy alone (n=792)
Primary tumour location at initial diagnosis				
Gastric cancer	333 (70%)	334 (69%)	554 (70%)	556 (70%)
Gastro-oesophageal junction cancer	84 (18%)	86 (18%)	132 (17%)	128 (16%)
Oesophageal adenocarcinoma	56 (12%)	62 (13%)	103 (13%)	108 (14%)
Microsatellite instability status				
Microsatellite stable	423 (89%)	423 (88%)	695 (88%)	682 (86%)
Microsatellite instability-high	18 (4%)	16 (3%)	23 (3%)	21 (3%)
Not reported or invalid	32 (7%)	43 (9%)	71 (9%)	89 (11%)

Overall survival

CPS ≥5



CPS ≥1



All patients

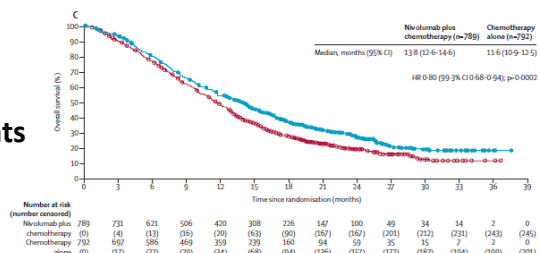
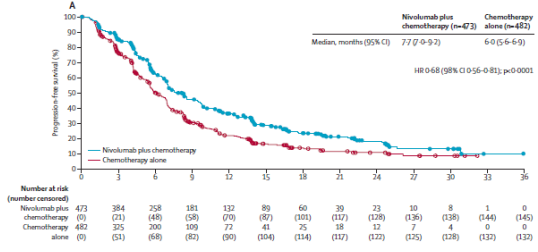


Table S3: Objective and best overall response per blinded independent central review

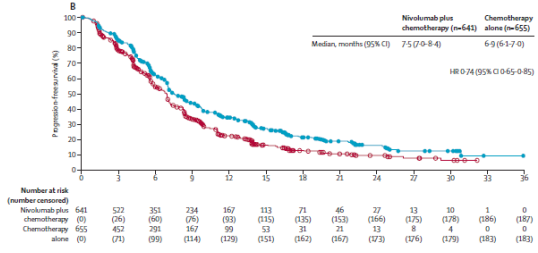
	Patients with PD-L1 CPS ≥5	
	Nivolumab plus chemotherapy (n=378)*	Chemotherapy alone (n=391)*
Objective response rate	226 (60%)	177 (45%)
95% CI	55–65	40–50
Best overall response†		
Complete response	44 (12%)	27 (7%)
Partial response	182 (48%)	150 (38%)
Stable disease	104 (28%)	132 (34%)
Progressive disease	26 (7%)	42 (11%)
Not evaluable	22 (6%)	40 (10%)

Progression free survival

CPS ≥5



CPS ≥1



All patients

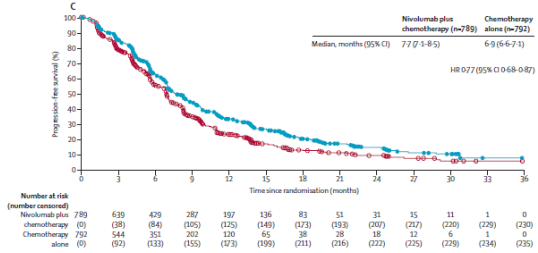


Figure S1: Subgroup analysis by PD-L1 CPS subpopulations

Population*	Median overall survival, months		Unstratified hazard ratio for death (95% CI)	Interaction test p value
	Nivolumab plus chemotherapy	Chemotherapy alone		
Overall (N=1581)	13.8	11.6	0.79 (0.70–0.89)	
PD-L1 CPS <1 (n=265)	13.1	12.5	0.92 (0.70–1.23)	
PD-L1 CPS ≥1 (n=1296)	14.0	11.3	0.76 (0.67–0.87)	0.2041
PD-L1 CPS <5 (n=606)	12.4	12.3	0.94 (0.78–1.13)	
PD-L1 CPS ≥5 (n=955)	14.4	11.1	0.70 (0.60–0.81)	0.0107†

B Progression-free survival

Population*	Median progression-free survival, months		Unstratified hazard ratio for progression/death (95% CI)	Interaction test p value
	Nivolumab plus chemotherapy	Chemotherapy alone		
Overall (N=1581)	7.7	6.9	0.77 (0.68–0.87)	
PD-L1 CPS <1 (n=265)	8.7	8.1	0.93 (0.69–1.26)	
PD-L1 CPS ≥1 (n=1296)	7.5	6.9	0.75 (0.65–0.85)	0.1391†
PD-L1 CPS <5 (n=606)	7.5	8.2	0.93 (0.76–1.12)	
PD-L1 CPS ≥5 (n=955)	7.7	6.1	0.69 (0.59–0.80)	0.0073†

C Objective response rate

Population‡§	Objective response rate, %		Unweighted objective response rate difference, % (95% CI)
	Nivolumab plus chemotherapy	Chemotherapy alone	
Overall (N=1211)	58	46	12 (6–17.5)
PD-L1 CPS <1 (n=178)	51	41	9 (–5 to 23)
PD-L1 CPS ≥1 (n=1019)	60	46	13 (7–19)
PD-L1 CPS <5 (n=428)	55	46	9 (–0.6 to 18)
PD-L1 CPS ≥5 (n=769)	60	45	15 (7.5–21)

CheckMate 649

	Nivolumab plus chemotherapy (n=782)*				Chemotherapy alone (n=767)*			
	Grade 1-2	Grade 3	Grade 4	Grade 5†	Grade 1-2	Grade 3	Grade 4	Grade 5
All events	272 (35%)	358 (46%)	104 (13%)	4 (1%)	338 (44%)	285 (37%)	56 (7%)	0
Serious events	37 (5%)	97 (12%)	34 (4%)	4 (1%)	16 (2%)	63 (8%)	14 (2%)	0
Events leading to discontinuation	148 (19%)	109 (14%)	23 (3%)	4 (1%)	114 (15%)	58 (8%)	9 (1%)	0
Any-grade events in 10% or more of treated patients in either group								
Nausea	303 (39%)	20 (3%)	0	0	273 (36%)	19 (2%)	0	0
Diarrhoea	218 (28%)	33 (4%)	2 (<1%)	0	182 (24%)	23 (3%)	1 (<1%)	0
Peripheral neuropathy	190 (24%)	29 (4%)	2 (<1%)	0	168 (22%)	22 (3%)	0	0
Vomiting	178 (23%)	17 (2%)	0	0	142 (19%)	24 (3%)	0	0
Fatigue	172 (22%)	30 (4%)	0	0	156 (20%)	16 (2%)	1 (<1%)	0
Anaemia	156 (20%)	44 (6%)	3 (<1%)	0	150 (20%)	20 (3%)	1 (<1%)	0
Decreased appetite	143 (18%)	14 (2%)	0	0	126 (16%)	12 (2%)	1 (<1%)	0
Thrombocytopenia	138 (18%)	15 (2%)	4 (1%)	0	132 (17%)	12 (2%)	1 (<1%)	0
Platelet count decreased	136 (17%)	17 (2%)	3 (<1%)	0	96 (13%)	15 (2%)	4 (1%)	0
Peripheral sensory neuropathy	121 (15%)	16 (2%)	0	0	105 (14%)	14 (2%)	0	0
Aspartate aminotransferase increased	110 (14%)	12 (2%)	0	0	64 (8%)	5 (1%)	0	0
White blood cell count decreased	89 (11%)	20 (3%)	3 (<1%)	0	64 (8%)	12 (2%)	1 (<1%)	0
Alanine aminotransferase increased	83 (11%)	6 (1%)	0	0	45 (6%)	5 (1%)	0	0
Palmar-plantar erythrodysesthesia syndrome	83 (11%)	11 (1%)	0	0	75 (10%)	6 (1%)	0	0
Neutrophil count decreased	75 (10%)	60 (8%)	23 (3%)	0	51 (7%)	50 (7%)	17 (2%)	0
Neutropenia	73 (9%)	87 (11%)	31 (4%)	0	88 (11%)	70 (9%)	23 (3%)	0
Asthenia	66 (8%)	7 (1%)	0	0	71 (9%)	9 (1%)	1 (<1%)	0
Lipase increased	44 (6%)	34 (4%)	11 (1%)	0	18 (2%)	14 (2%)	2 (<1%)	0

	ChT + Nivo	ChT alone
G3-4 AEs	59%	44%
AEs leading to discontinuation	36%	24%
SAEs	22%	12%

KEYNOTE-590

Randomised, placebo-controlled, double-blind, phase 3 study.

Stratified by geographical region (Asia vs non-Asia), histology (oesophageal squamous cell carcinoma vs adenocarcinoma), and ECOG performance status (0 vs 1).

Outcomes: dual primary endpoints were OS and PFS in patients with oesophageal SCC and CPS ≥10 (not for PFS), oesophageal SCC, CPS ≥10 and all randomized patients.

Adult patients with untreated, locally advanced, unresectable or metastatic adenocarcinoma or squamous cell carcinoma of the oesophagus or Siewert type 1 GEJ adenocarcinoma
(n = 749)

Randomized 1:1

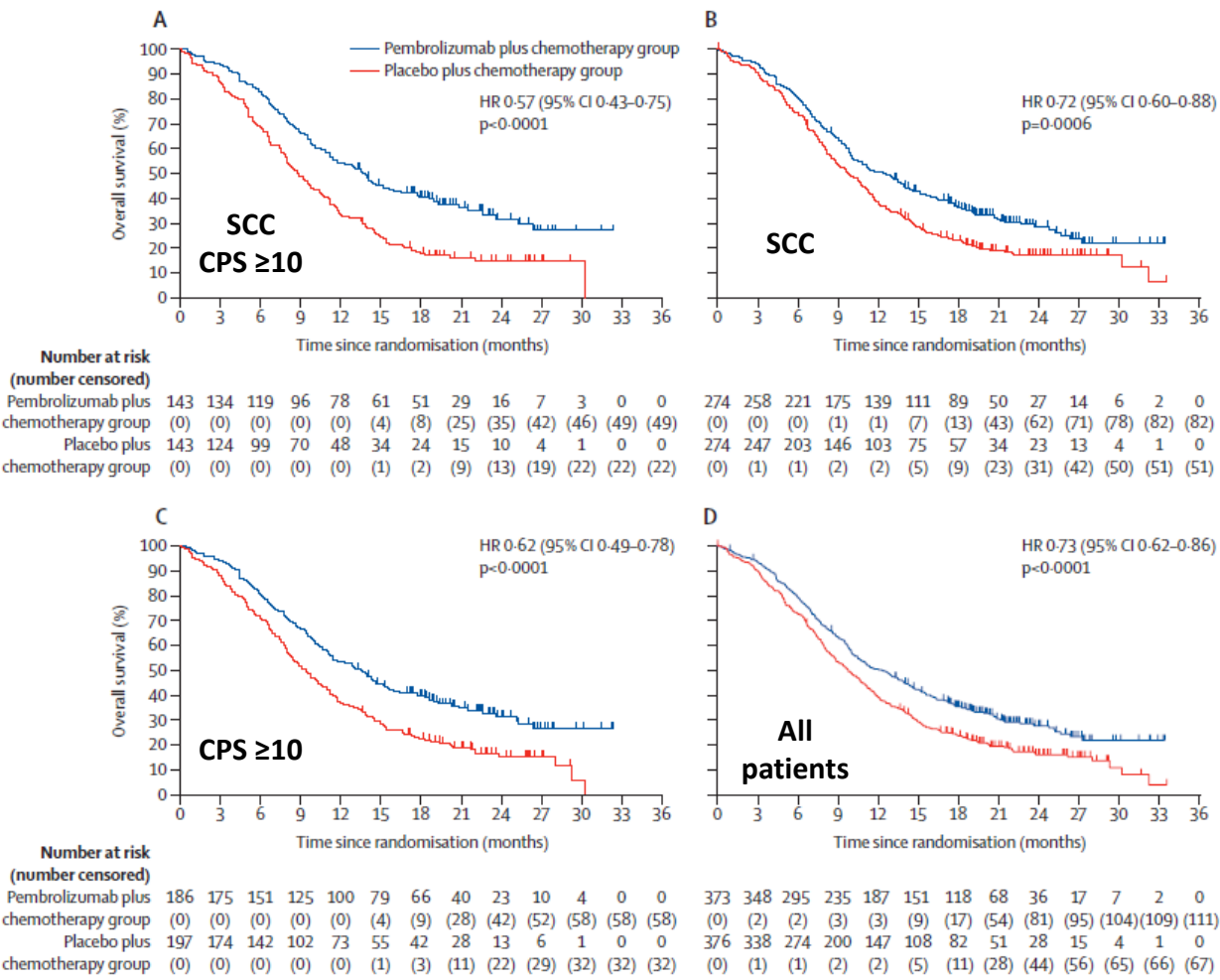
Pembrolizumab 200 mg Q3W up to 35 cycles +
Chemotherapy (FP) Q3W up to 6 cycles
(n = 373)

Placebo Q3W +
Chemotherapy (FP) Q3W up to 6 cycles
(n = 376)

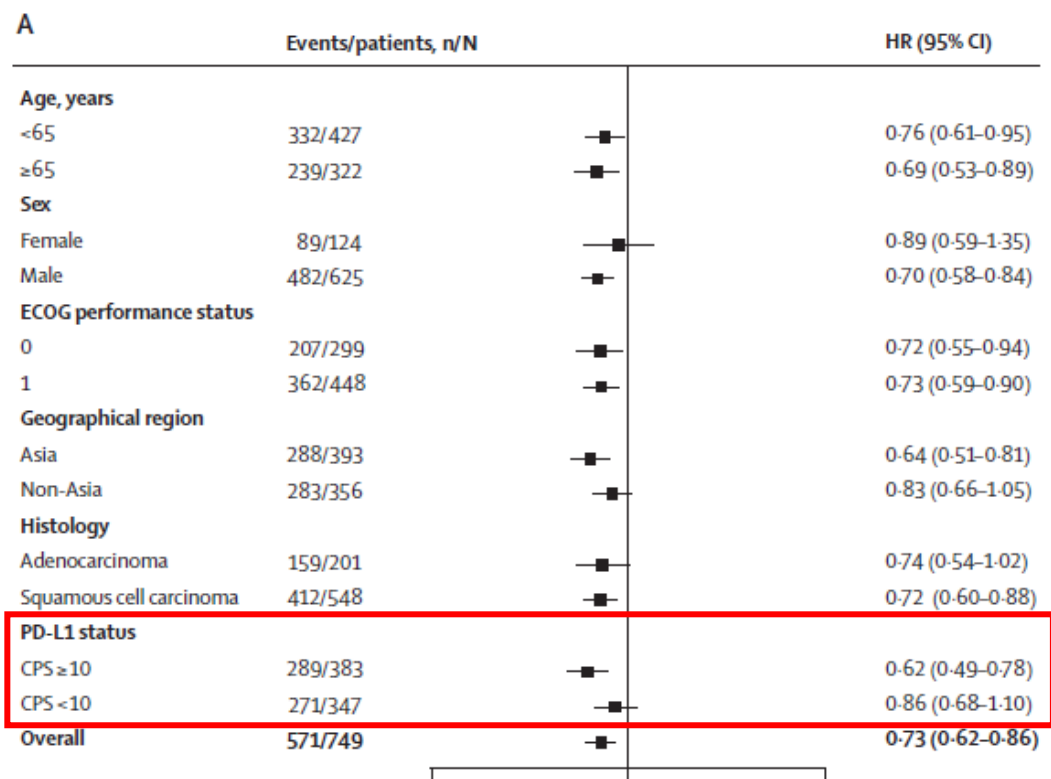
	Pembrolizumab plus chemotherapy group (n=373)	Placebo plus chemotherapy group (n=376)
Oesophageal squamous cell carcinoma	274 (73%)	274 (73%)
Adenocarcinoma	99 (27%)	102 (27%)
Oesophageal adenocarcinoma	58 (16%)	52 (14%)
Siewert type 1 gastro- oesophageal junction adenocarcinoma‡	41 (11%)	50 (13%)

KEYNOTE-590

Overall survival



Overall survival



	ChT + Pembro	ChT alone
Any AEs	100%	99%
G3-4 AEs	86%	83%
AEs leading to discontinuation	24%	20%
irAEs	26%	12%

	Pembrolizumab plus chemotherapy group (n=370)		Placebo plus chemotherapy group (n=370)	
	Any	Grade ≥3	Any	Grade ≥3
(Continued from previous page)				
Adverse events of special interest†				
Hypothyroidism	40 (11%)	0	24 (6%)	0
Pneumonitis	23 (6%)	2 (1%)	2 (1%)	1 (<1%)
Hyperthyroidism	21 (6%)	1 (<1%)	3 (1%)	0
Colitis	8 (2%)	4 (1%)	6 (2%)	3 (1%)
Infusion reactions	6 (2%)	1 (<1%)	4 (1%)	0
Hepatitis	5 (1%)	5 (1%)	0	0
Adrenal insufficiency	4 (1%)	2 (1%)	2 (1%)	0
Severe skin reactions	4 (1%)	4 (1%)	2 (1%)	2 (1%)
Hypophysitis	3 (1%)	1 (<1%)	0	0
Pancreatitis	2 (1%)	0	1 (<1%)	1 (<1%)
Myositis	1 (<1%)	1 (<1%)	0	0
Nephritis	1 (<1%)	0	2 (1%)	1 (<1%)
Thyroiditis	1 (<1%)	0	0	0
Type 1 diabetes	1 (<1%)	1 (<1%)	0	0

KEYNOTE-859

Multicentre, double-blind, placebo-controlled, randomised, phase 3 trial.

Stratified by geographical region (western Europe, Israel, North America, and Australia vs Asia vs rest of world), PD-L1 CPS (<1 vs ≥1), and investigator’s choice of chemotherapy (FP vs CAPOX).

Outcomes: primary endpoint OS.

Adult patients with untreated unresectable LA or metastatic HER2-negative adenocarcinoma of the stomach or GEJ
(n = 1579)



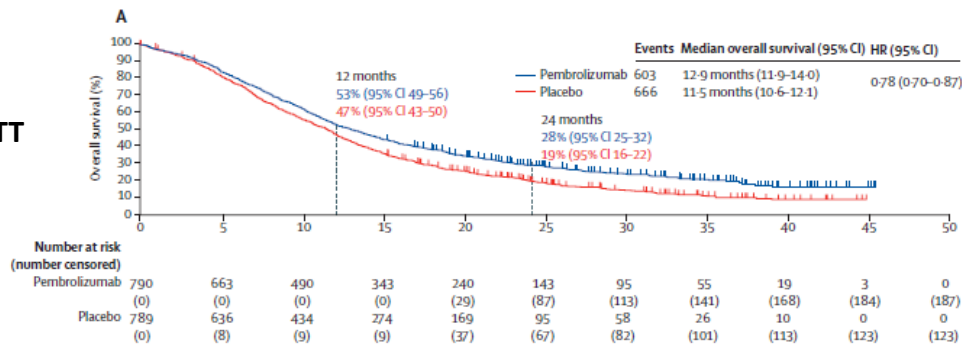
Pembrolizumab 200 mg Q3W up to 35 cycles +
Chemotherapy (FP or CAPOX) Q3W up to 6 cycles*
(n = 790)

Placebo Q3W +
Chemotherapy (FP or CAPOX) Q3W up to 6 cycles*
(n = 789)

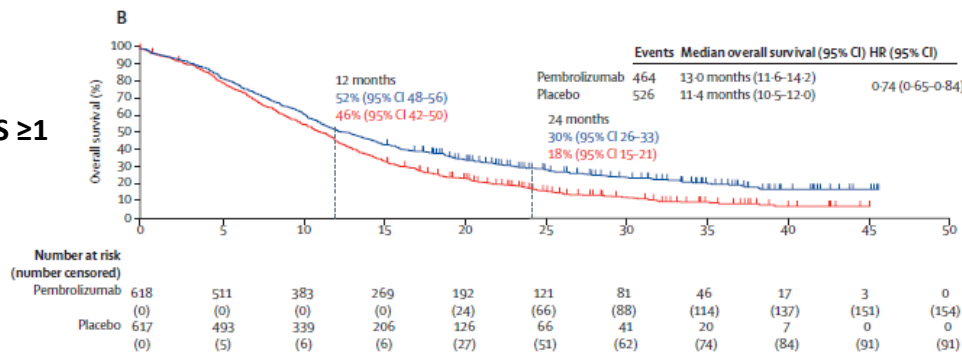
	Pembrolizumab plus chemotherapy (n=790)			Placebo plus chemotherapy (n=789)		
	ITT (n=790)	PD-L1 CPS ≥1 (n=618)	PD-L1 CPS ≥10 (n=279)	ITT (n=789)	PD-L1 CPS ≥1 (n=617)	PD-L1 CPS ≥10 (n=272)
Primary tumour location						
Gastro-esophageal junction	149 (19%)	123 (20%)	65 (23%)	185 (23%)	164 (27%)	73 (27%)
Stomach	640 (81%)	494 (80%)	214 (77%)	603 (76%)	453 (73%)	199 (73%)
Other	0	0	0	1 (<1%)	0	0
Missing	1 (<1%)	1 (<1%)	0	0	0	0
Microsatellite instability status						
High	39 (5%)	35 (6%)	20 (7%)	35 (4%)	31 (5%)	16 (6%)
Low or microsatellite stable	641 (81%)	503 (81%)	227 (81%)	639 (81%)	500 (81%)	224 (82%)
Unknown	0	0	0	1 (<1%)	1 (<1%)	1 (<1%)
PD-L1 CPS						
≥1	618 (78%)	618 (100%)	279 (100%)	617 (78%)	617 (100%)	272 (100%)
<1	172 (22%)	0	0	172 (22%)	0	0
≥10	279 (35%)	279 (45%)	279 (100%)	272 (34%)	272 (44%)	272 (100%)
<10	509 (64%)	337 (55%)	0	517 (66%)	345 (56%)	0
Missing	2 (<1%)	2 (<1%)	0	0	0	0

KEYNOTE-859

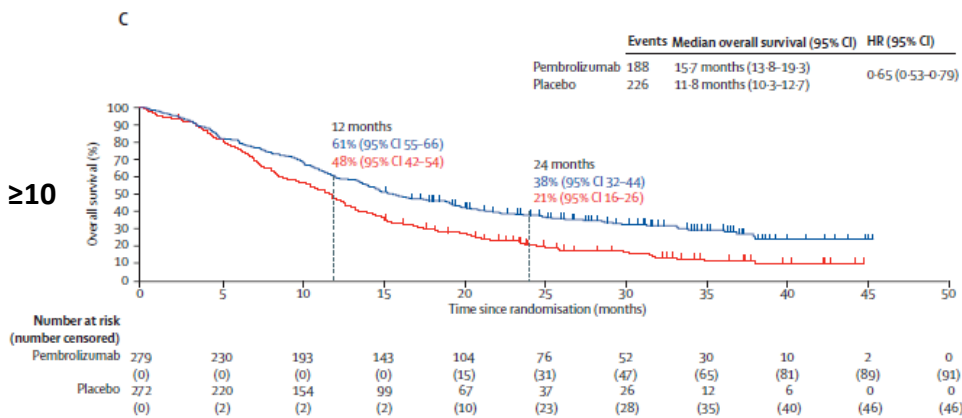
ITT



CPS ≥1



CPS ≥10



	Pembrolizumab group, n/N	Placebo group, n/N	HR (95% CI)
Overall	603/790	666/789	0.78 (0.70-0.87)
Age, years			
<65	383/486	416/479	0.76 (0.66-0.88)
≥65	220/304	250/310	0.77 (0.64-0.92)
Sex			
Female	208/263	218/245	0.76 (0.63-0.92)
Male	395/527	448/544	0.77 (0.67-0.88)
Race			
Asian	190/270	212/269	0.72 (0.59-0.88)
Non-Asian	408/513	447/512	0.80 (0.70-0.92)
Geographical location			
Asia	184/263	206/262	0.71 (0.58-0.87)
Western Europe, Israel, North America, and Australia	160/201	172/202	0.80 (0.65-1.00)
Rest of the world	259/326	288/325	0.80 (0.68-0.95)
ECOG performance status			
0	193/281	246/301	0.73 (0.60-0.88)
1	410/509	420/488	0.77 (0.68-0.89)
Primary tumour location			
Gastro-esophageal junction	117/149	159/185	0.74 (0.58-0.94)
Stomach	485/640	507/603	0.77 (0.68-0.87)
Histological subtype			
Diffuse	259/318	269/301	0.76 (0.64-0.90)
Indeterminate	132/186	177/215	0.69 (0.55-0.87)
Intestinal	210/284	220/273	0.81 (0.67-0.98)
Disease status			
Metastatic	582/761	643/759	0.77 (0.69-0.86)
Liver metastases			
No	355/475	401/478	0.73 (0.63-0.84)
Yes	247/314	265/311	0.83 (0.70-0.99)
Previous gastrectomy or oesophagectomy			
No	487/613	535/622	0.79 (0.70-0.90)
Yes	112/172	126/162	0.69 (0.54-0.90)
Microsatellite instability status			
High	14/39	25/35	0.34 (0.18-0.66)
Low or microsatellite stable	497/641	540/639	0.79 (0.70-0.89)
PD-L1 CPS at baseline			
<1	139/172	140/172	0.92 (0.73-1.17)
≥1	464/618	526/617	0.73 (0.65-0.83)
1-9	274/337	300/345	0.83 (0.70-0.98)
<10	413/509	440/517	0.86 (0.75-0.98)
≥10	188/279	226/272	0.64 (0.52-0.77)
Chemotherapy choice at randomisation			
Capecitabine and oxaliplatin	509/682	567/681	0.76 (0.68-0.86)
Fluorouracil and cisplatin	94/108	99/108	0.82 (0.62-1.09)

0.10.31.03.0

Favours pembrolizumabFavours placebo

KEYNOTE-859

	Pembrolizumab plus chemotherapy (n=785)				Placebo plus chemotherapy (n=787)			
	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
Any event	285 (36%)	405 (52%)	53 (7%)	8 (1%)*	334 (42%)	352 (45%)	34 (4%)	16 (2%)†
Nausea	299 (38%)	25 (3%)	1 (<1%)	0	297 (38%)	29 (4%)	0	0
Diarrhoea	206 (26%)	42 (5%)	3 (<1%)	1 (<1%)	177 (22%)	35 (4%)	1 (<1%)	1 (<1%)
Anaemia	179 (23%)	62 (8%)	2 (<1%)	0	161 (20%)	48 (6%)	3 (<1%)	0
Vomiting	180 (23%)	35 (4%)	0	0	143 (18%)	32 (4%)	0	0
Decreased platelet count	141 (18%)	47 (6%)	8 (1%)	0	141 (18%)	32 (4%)	4 (1%)	0
Decreased neutrophil count	121 (15%)	63 (8%)	9 (1%)	0	112 (14%)	54 (7%)	4 (1%)	0
Palmar-plantar erythrodysaesthesia syndrome	165 (21%)	24 (3%)	0	0	152 (19%)	14 (2%)	0	0
Decreased appetite	153 (19%)	15 (2%)	0	0	154 (20%)	14 (2%)	0	0
Fatigue	130 (17%)	27 (3%)	0	0	132 (17%)	32 (4%)	0	0
Peripheral neuropathy	140 (18%)	10 (1%)	0	0	139 (18%)	24 (3%)	1 (<1%)	0
Neutropenia	87 (11%)	50 (6%)	5 (1%)	0	75 (10%)	52 (7%)	8 (1%)	0
Increased aspartate aminotransferase	128 (16%)	11 (1%)	0	0	94 (12%)	7 (1%)	1 (<1%)	0
Peripheral sensory neuropathy	115 (15%)	22 (3%)	0	0	123 (16%)	8 (1%)	0	0
Hypothyroidism	106 (14%)	1 (<1%)	0	0	32 (4%)	0	0	0
Increased alanine aminotransferase	91 (12%)	10 (1%)	0	0	61 (8%)	7 (1%)	0	0
Decreased white blood cell count	89 (11%)	9 (1%)	3 (<1%)	0	78 (10%)	7 (1%)	2 (<1%)	0
Asthenia	81 (10%)	12 (2%)	1 (<1%)	0	63 (8%)	16 (2%)	0	0
Thrombocytopenia	71 (9%)	10 (1%)	2 (<1%)	0	59 (7%)	18 (2%)	0	0
Increased blood bilirubin	69 (9%)	8 (1%)	1 (<1%)	0	48 (6%)	3 (<1%)	0	0

	Pembrolizumab plus chemotherapy (n=785)				Placebo plus chemotherapy (n=787)			
	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
Any	151 (19%)	57 (4%)	4 (1%)	1 (<1%)	60 (8%)	11 (1%)	1 (<1%)	1 (<1%)
Hypothyroidism	119 (15%)	1 (<1%)	0	0	34 (4%)	0	0	0
Hyperthyroidism	44 (6%)	0	0	0	13 (2%)	0	0	0
Colitis	7 (1%)	18 (2%)	1 (<1%)	0	10 (1%)	4 (1%)	0	0
Pneumonitis	15 (2%)	8 (1%)	1 (<1%)	1 (<1%)	4 (1%)	2 (<1%)	0	1 (<1%)
Thyroiditis	9 (1%)	0	0	0	1 (<1%)	0	0	0
Adrenal insufficiency	6 (1%)	3 (<1%)	1 (<1%)	0	1 (<1%)	0	0	0
Hepatitis	6 (1%)	3 (<1%)	0	0	1 (<1%)	2 (<1%)	1 (<1%)	0
Severe skin reactions	4 (1%)	12 (2%)	0	0	0	1 (<1%)	0	0
Hypophysitis	2 (<1%)	1 (<1%)	0	0	0	0	0	0
Vasculitis	1 (<1%)	1 (<1%)	0	0	1 (<1%)	0	0	0
Hypoparathyroidism	1 (<1%)	0	0	0	0	0	0	0
Uveitis	1 (<1%)	0	0	0	0	0	0	0
Myositis	0	1 (<1%)	0	0	0	0	0	0
Myasthenic syndrome	0	1 (<1%)	0	0	0	0	0	0
Myocarditis	0	0	0	0	0	1 (<1%)	0	0
Nephritis	0	4 (1%)	0	0	0	0	0	0
Pancreatitis	0	3 (<1%)	0	0	2 (<1%)	1 (<1%)	0	0
Type 1 diabetes	0	4 (1%)	1 (<1%)	0	1 (<1%)	0	0	0

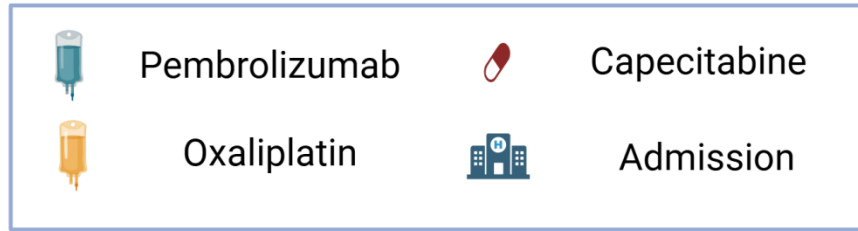
	ChT + pembro	ChT alone
Any AEs	99%	98%
GR3-5 AEs	8%	2%
SAEs	23%	19%
AEs leading to discontinuation	26%	20%
irAEs	29%	7%

61 year old previously healthy male diagnosed with **MMRp, CPS 18, HER2-negative, claudin18.2-negative stage IV** (non-locoregional lymph nodes) **adenocarcinoma** of the **distal esophagus**



Due to high PDL1 expression (CPS) our patient started **first-line** systemic treatment with **CAPOX + pembrolizumab**

First-line treatment timeline



1



2



3



4



Hospital admission:

- Grade 3 vomiting.
- Stage 2 AKI (creatinine 2,87 mg/dL; baseline 1)

Fist day after admission presents fever and grade 3 diarrhea (10 stools/day)

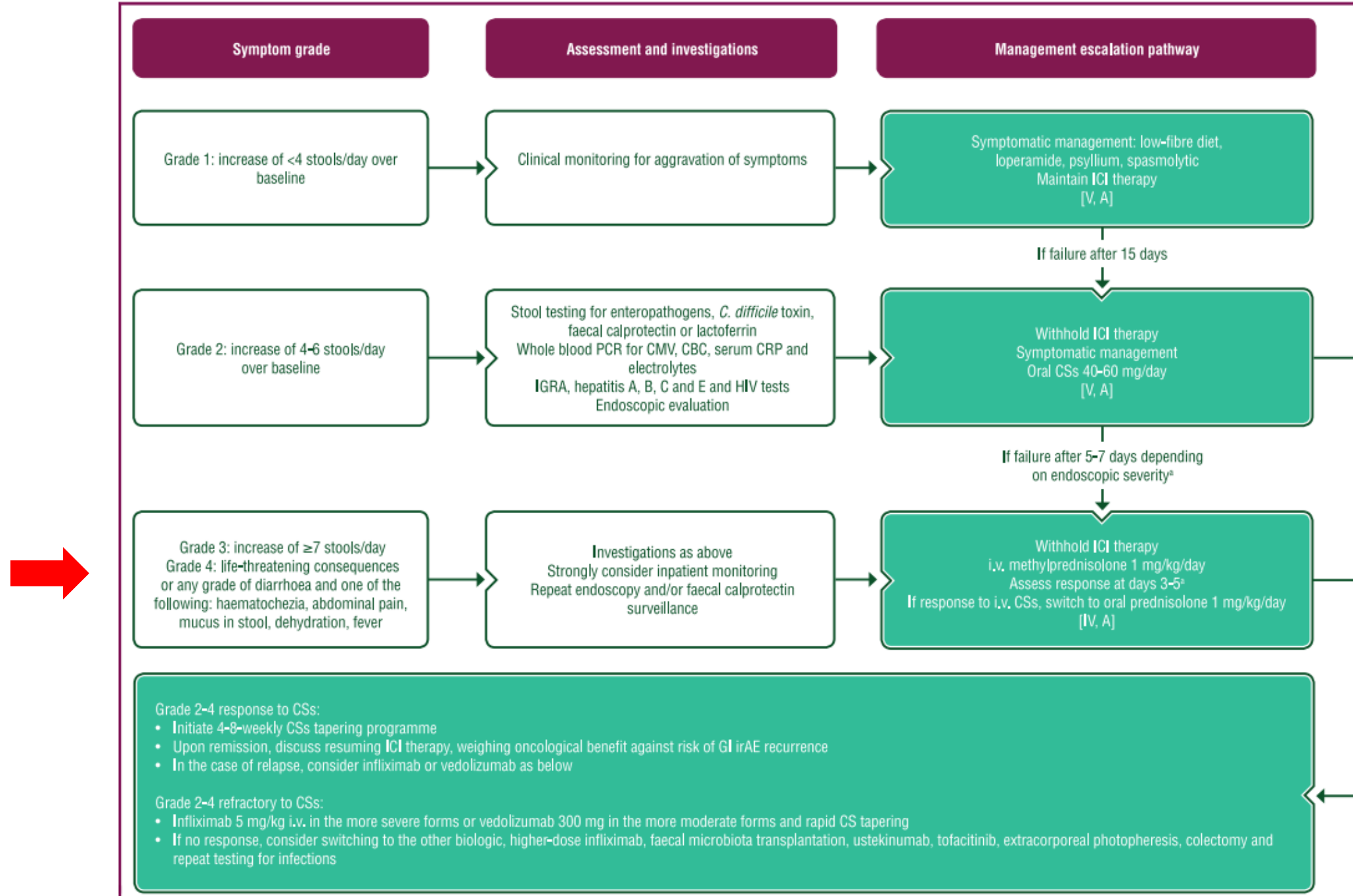


Figure 5. Management of IR-diarrhoea and enterocolitis.

First-line treatment timeline



Pembrolizumab



Capecitabine



Oxaliplatin



Admission

1



2



3



4



Hospital admission:

- Grade 3 vomiting.
- Stage 2 AKI (creatinine 2,87 mg/dL; baseline 1)

Fist day after admission presents fever and grade 3 diarrhea (10 stools/day)



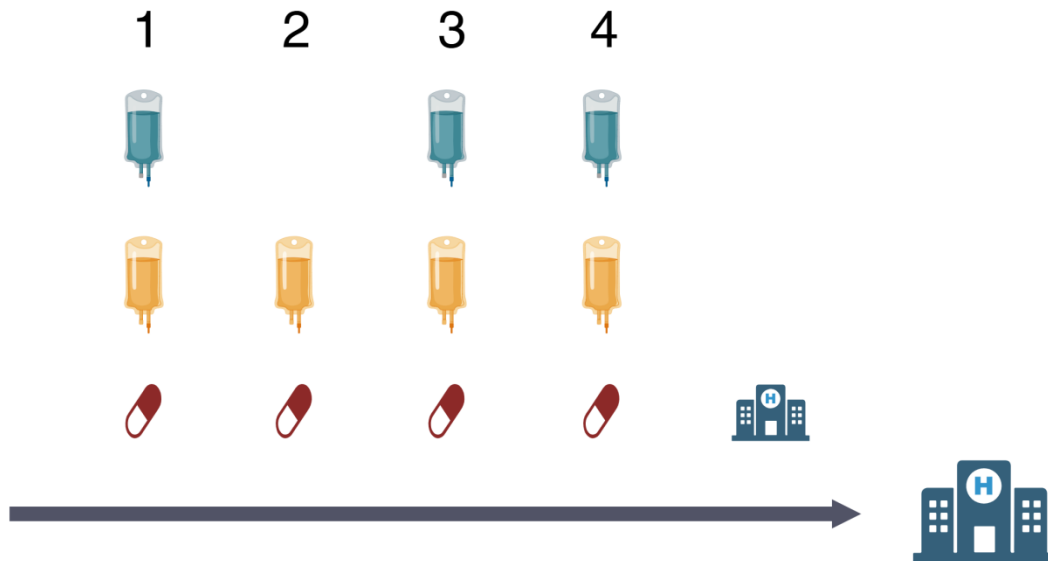
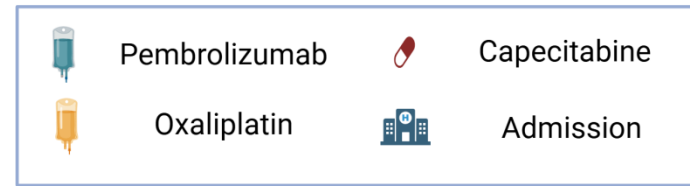
¿IR colitis?

Methylprednisolone 1mg/kg



- Blood & stool culture: negative.
- Rectal endoscopy: normal.
- Bowel rythm normalized.
- Creatinine: 1,23 mg/dL.

First-line treatment timeline



Hospital admission:

- Fever.
- Stage 2 AKI (creatinine 3,01 mg/dL)

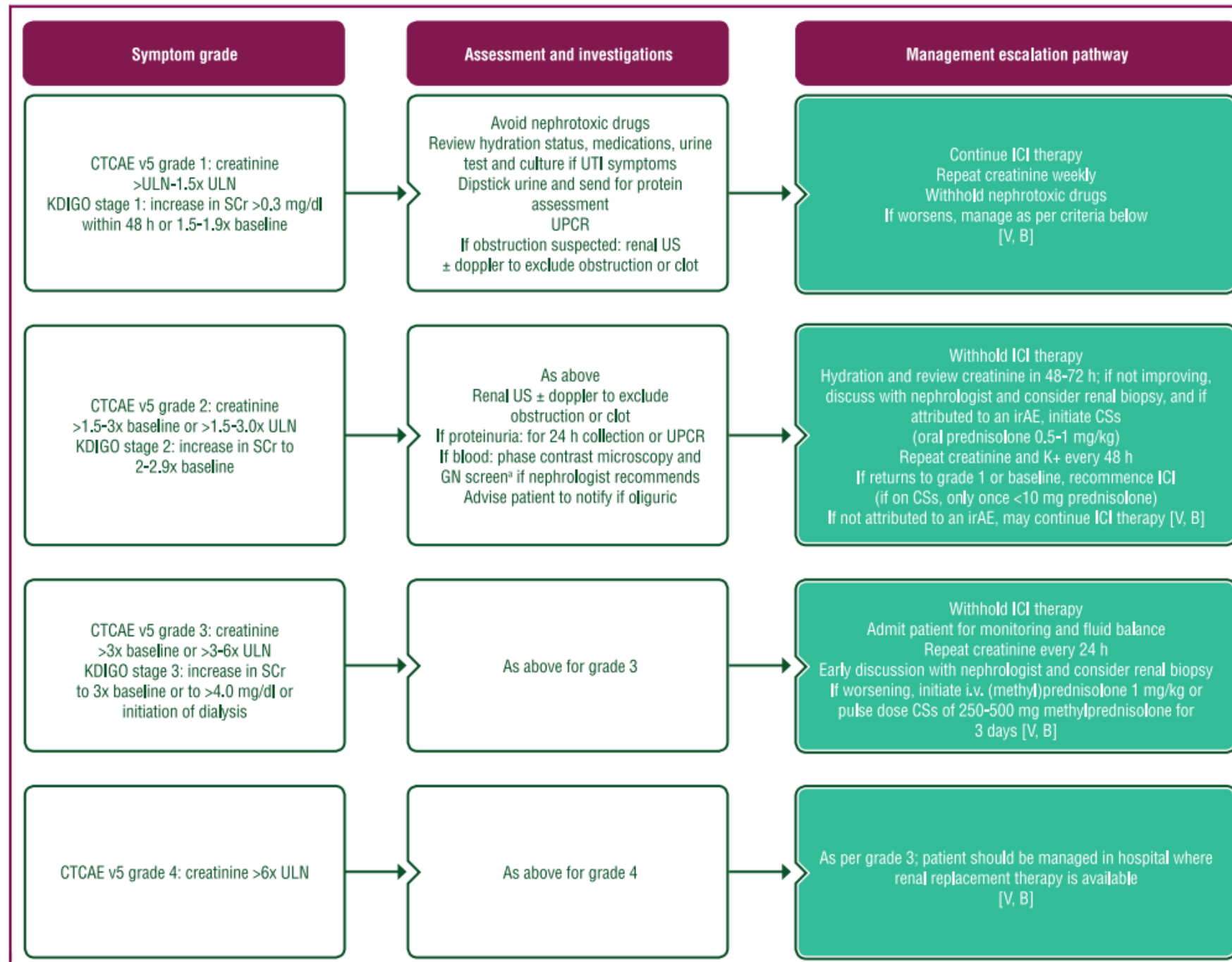
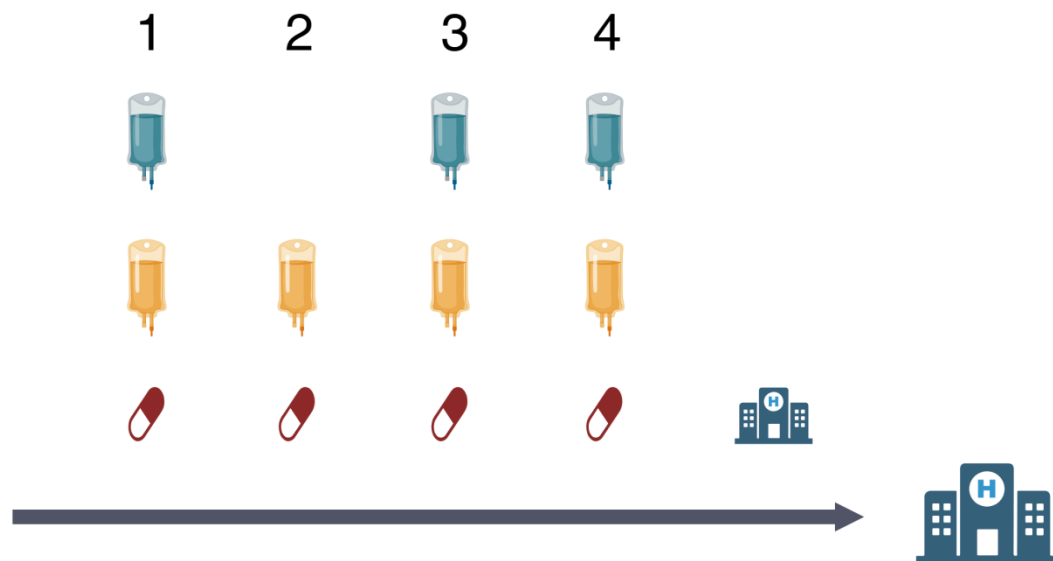
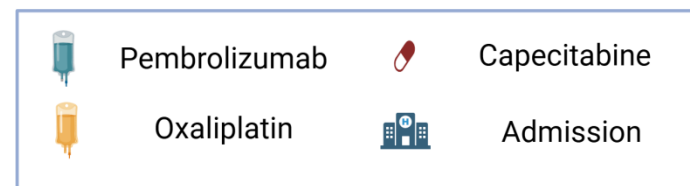


Figure 10. Management of IR-renal toxicity.

First-line treatment timeline



Hospital admission:

- Fever.
- Stage 2 AKI (creatinine 3,01 mg/dL)



Renal US: normal. No hydronephrosis.



¿IR nephritis?

Methylprednisolone 1mg/kg
Discussion with Nephrology



- Urinalysis compatible with IR nephritis + good response to corticosteroid treatment (creat 1,77).
- **RENAL BIOPSY AVOIDED**
- Hospital discharge. **Ambulatory corticosteroid tapering.**

First-line treatment timeline



Pembrolizumab



Capecitabine



Oxaliplatin



Admission

1



2



3



4



5



↓
Dose
red.



↓
Dose
red.



Grade 3 nephritis

Take home messages

- Biomarker testing is key in order to adequately select the first-line treatment for advanced gastroesophageal adenocarcinoma.
- The addition of immunotherapy to chemotherapy improves outcomes in patients whose tumors highly express PDL1.
- The addition of immunotherapy may lead to the development of immune-related adverse events (irAEs) which can affect all organs.
- Immune-related toxicity should be monitored prior to immunotherapy administration. irAEs should be managed according to its grade (see ESMO Clinical Practice Guidelines).
- Grade 1-2 irAEs can be managed in the outpatient setting and its treatment often implies surveillance $\pm$ low dose oral corticosteroids.
- Grade 3-4 irAEs management includes strict surveillance $\pm$ hospital admission, high dose corticosteroids and may require immunosuppressants. Specialist consultation is advised.